

DATA INTERPRETATION

[Ratio, Average, Percentage, Profit & Loss]

RATIO :

Ratio means division ($\div$)

The ratio of a, b is a : b

$$a : b \neq b : a$$

The terms of ratio must be in simplified form.

Ex : i) $120 : 160 : 200 = ?$

$$120 : 160 : 200 = 3 : 4 : 5 \quad [\div 40]$$

ii) $85 : 119 : 153 = ?$

$$85 : 119 : 153 = 5 : 7 : 9 \quad [\div 17]$$

AVERAGE :

$$\text{Average} = \frac{\text{Sum}}{\text{Number}}$$

Ex : Average of (37, 53, 24, 16, 40) is ?

$$\text{Average} = \frac{\text{Sum}}{\text{Number}} = \frac{37+53+24+16+40}{5} = \frac{170}{5} = 34$$

PERCENTAGE:

- The value is increased / decreased from P to Q, then it's

$$\% = \frac{(P - Q) \text{ or } (Q - P)}{P} \times 100\%$$

- P is how much % more / less than Q ----- $\frac{(P - Q) \text{ or } (Q - P)}{Q} \times 100\%$

- P is what % of Q ----- $P = \frac{X}{100} \times Q$

- 23456 ----- 10 % = 2345.6

$$1\% = 234.56$$

$$0.1\% = 23.456$$

Ex : In 4000 ----- 10% = 400

$$1\% = 40$$

$$0.1\% = 4$$

How can you calculate 5%, 20%, 30%, 25%, 50% & 75% :

$$\text{Base} = 10\% \text{ ----- } 5\% = \frac{10\%}{2}$$

$$20\% = 10\% \times 2$$

EDUDARTS

MNCs (TCS, Infosys,) - APTITUDE

CAMPUS
PLACEMENT
SKILLS

$$30\% = 10\% \times 3$$

$$25\% = \frac{1}{4} \quad ; \quad 50\% = \frac{1}{2} \quad ; \quad 75\% = \frac{3}{4}$$

PERCENTAGE MEMORY SHEET (Total = 100%)		
Fraction	Percentage	Decimal Value
1/2	50%	0.5
1/3	33.33%	0.3333
1/4	25%	0.25
1/5	20%	0.2
1/6	16.66%	0.1666
1/7	14.28%	0.1428
1/8	12.5%	0.125
1/9	11.11%	0.1111
1/10	10%	0.1
1/11	09.09%	0.0909
1/12	8.33%	0.0833
1/13	7.69%	0.0769
1/14	7.14%	0.0714
1/15	6.66%	0.0666
1/16	6.25%	0.0625
1/17	5.88%	0.0588
1/18	5.55%	0.0555
1/19	5.26%	0.0526
1/20	5%	0.05
1/21	4.76%	0.0476
1/22	4.54%	0.0454
1/23	4.35%	0.0435
1/24	4.16%	0.0416
1/25	4%	0.04
1/26	3.84%	0.0384
1/27	3.7%	0.037
1/28	3.57%	0.0357
1/29	3.45%	0.0345
1/30	3.33%	0.0333

Ex : i) 10.4% of 234 = ?

ii) 20.3% of 345 = ?

iii) 25.2% of 468 = ?

iv) 50.1% of 624 = ?

v) 14.28% of 287 = ?

vi) 37.5% of 248 = ?

vii) 44.44% of 1818 = ?

viii) 45.45% of 121 = ?

Profit & Loss :

- $\text{Gain} = \text{S.P} - \text{C.P}$
- $\text{Loss} = \text{C.P} - \text{S.P}$
- $\text{Gain\%} = \frac{\text{Gain}}{\text{C.P}} \times 100\%$
- $\text{Loss\%} = \frac{\text{Loss}}{\text{C.P}} \times 100\%$
- $\text{S.P} = \frac{(100 + \text{G}\%)}{100} \times \text{C.P}$
- $\text{S.P} = \frac{(100 - \text{G}\%)}{100} \times \text{C.P}$

- $\text{C.P} = \frac{100}{(100 + \text{G}\%)} \times \text{S.P}$
- $\text{C.P} = \frac{100}{(100 - \text{L}\%)} \times \text{S.P}$
- Marked Price is always more than Cost Price
- Discount is offered only on Marked Price(M.P)
- $\text{Selling Price(S.P)} = \text{Marked Price} - \text{Discount}$

Tabular Graph

(1-5): Study the given table carefully to answer the following questions. The table provides Colleges, Courses and number of students. [Infosys]

Courses Colleges	B.Tech	M.Tech	MBA	MCA	Ph.D
P	100	80	40	30	10
Q	80	100	50	60	30
R	90	50	30	20	8
S	60	70	20	100	40
T	120	40	50	30	25

- All the colleges together, which course has least strength ?
a)M.Tech b)MBA c)Ph.D d)MCA
- Which college has highest strength ?
a)P b)Q c)S d)T
- The number of MBA Students in College Q is what percent of M.Tech Students in College S ?
a)68.3% b)74.5% c)825 d)71.4%
- What is the ratio of number of MBA Students to MCA Students ?
a)22 : 15 b)11 : 15 c)11 : 14 d)11 : 30
- The number of B.Tech students in S is how much% less than M.Tech students in college P ?
a)25% b)20% c)30% d)40%

“Coding+Aptitude+Reasoning+Soft Skills = MNC Job” — EDUDARTS

Directions (6-10) : Study the table carefully, answer the questions given below. [Capgemini]

The following table shows X Class pass % .

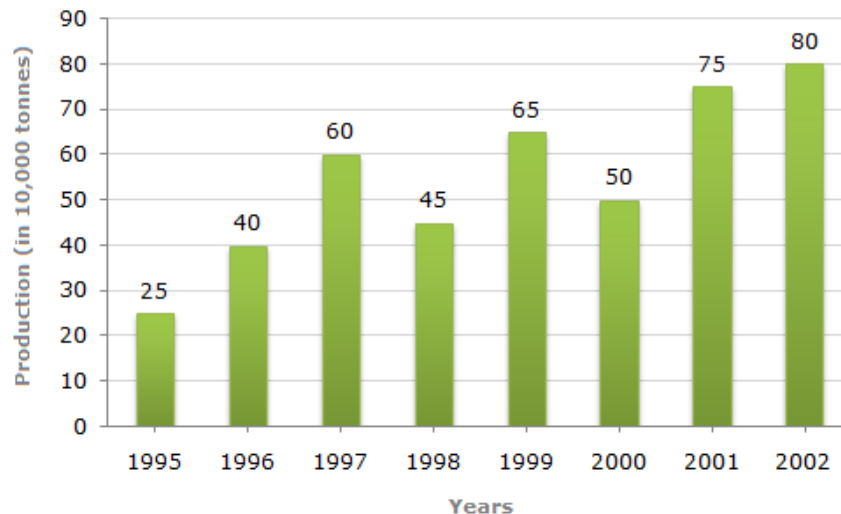
Year School	2011	2012	2013	2014	2015
Camford	40	60	50	80	30
Oxford	30	40	60	50	70
Cambridge	60	50	40	60	80
Nissan	50	30	70	40	90
Angels	70	80	40	30	60

6. What is the average pass % in 2011 ?
a) 50% b) 48% c) 60% d) can't be determined
7. 2011 to 2015, what is the average pass% of school Angels ?
a) 56% b) 54% c) 60% d) can't be determined
8. 800 students were attended for X Class in 2013, what is the average pass% ?
a) 52% b) 50% c) 60% d) can't be determined
9. 1200 students were pass in 2014 in X Class, what is the average pass % ?
a) 52% b) 48% c) 54% d) can't be determined
10. Which is the best school among 5 school ?
a) Angels b) Cambridge c) Nissan d) Camford

"Placement does not happen by luck, it happens only by practice"
- EDUDARTS

BAR GRAPH

Directions(1 -5) : Study the bar chart and answer the question based on it. [Cognizant]
Production of Fertilizers by a Company (in 10000 tonnes) Over the Years



1. What was the percentage decline in the production of fertilizers from 1997 to 1998?
a) 20% b) 15% c) 10% d) 25%
2. The average production of 1996 and 1997 was exactly equal to the average production of which of the following pairs of years?
a) 1995 & 2001 b) 2000 & 2001 c) 1999 & 2000 d) 1998 & 2000

3. What was the percentage increase in production of fertilizers in 2002 compared to that in 1995?

- a)100% b)200% c)220% d)180%

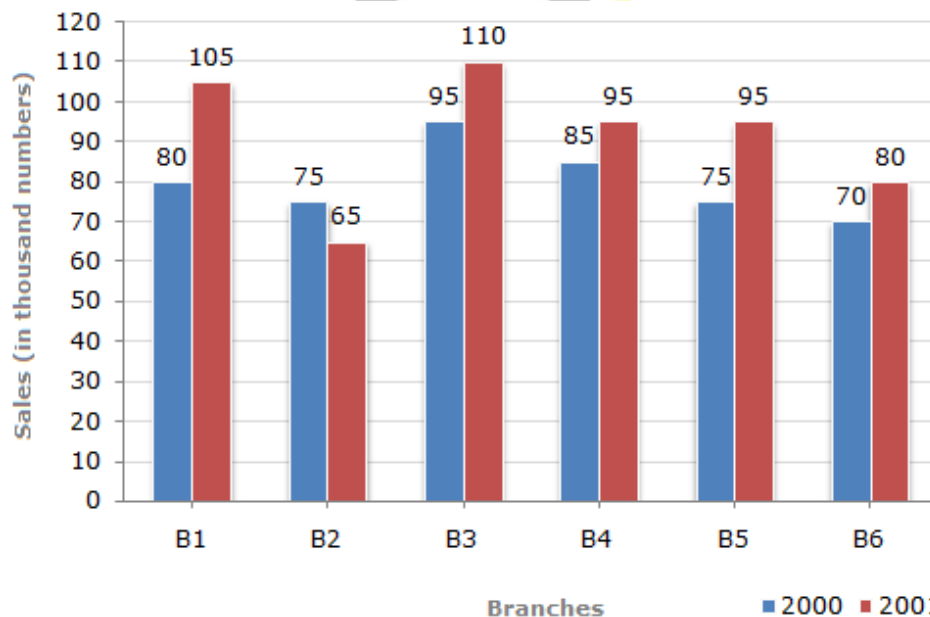
4. In which year was the percentage increase in production as compared to the previous year the maximum?

- a)1995 b)1996 c)2002 d)2001

5. In how many of the given years was the production of fertilizers more than the average production of the given years?

- a)1 b)2 c)3 d)4

Directions (6 - 10) : The bar graph given below shows the sales of books (in thousand number) from six branches of a publishing company during two consecutive years 2000 and 2001. Sales of Books (in thousand numbers) from Six Branches - B1, B2, B3, B4, B5 and B6 of a publishing Company in 2000 and 2001. [IBM]



6. What is the ratio of the total sales of branch B2 for both years to the total sales of branch B4 for both years?

- a)3 : 5 b)7 : 9 c)9 : 7 d)5 : 3

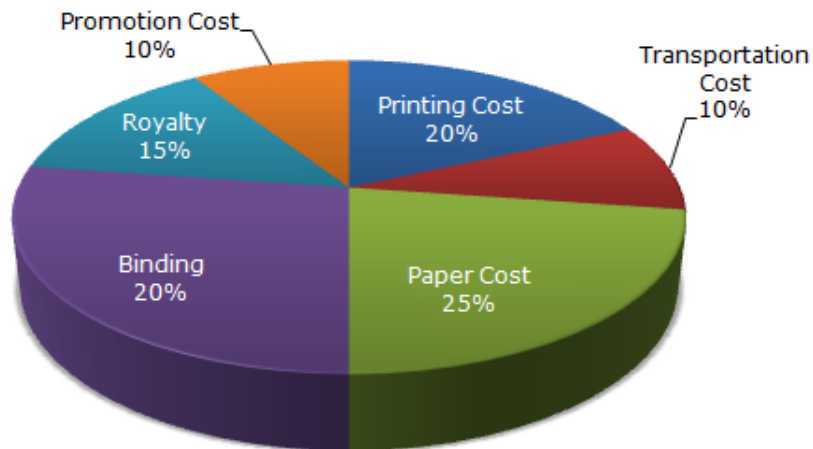
7. Total sales of branch B6 for both the years is what percent of the total sales of branches B3 for both the years?

- a)73.17% b)62.9% c)81.2% d)77.1%

8. What percent of the average sales of branches B1, B2 and B3 in 2001 is the average sales of branches B1, B3 and B6 in 2000?
a)46.5% b)57.5% c)87.5% d)90.7%
9. What is the average sales of all the branches (in thousand numbers) for the year 2000?
a)40 b)30 c)60 d)80
10. Total sales of branches B1, B3 and B5 together for both the years (in thousand numbers) is?
a)560 b)650 c)460 d)550

PIE - CHART

Directions (1-5) : The following pie-chart shows the percentage distribution of the expenditure incurred in publishing a book. Study the pie-chart and answer the questions based on it. **Various Expenditures (in percentage) Incurred in Publishing a Book [TCS]**



1. If for a certain quantity of books, the publisher has to pay ₹. 30,600 as printing cost, then what will be amount of royalty to be paid for these books?
a) ₹.22000 b) ₹.22950 c) ₹.22900 d) ₹.23000
2. What is the central angle of the sector corresponding to the expenditure incurred on Royalty?
a) 54° b) 36° c) 32° d) 72°
3. The price of the book is marked 20% above the C.P. If the marked price of the book is ₹.180, then what is the cost of the paper used in a single copy of the book?
a) ₹.34 b) ₹.35 c) ₹.37.5 d) ₹.40

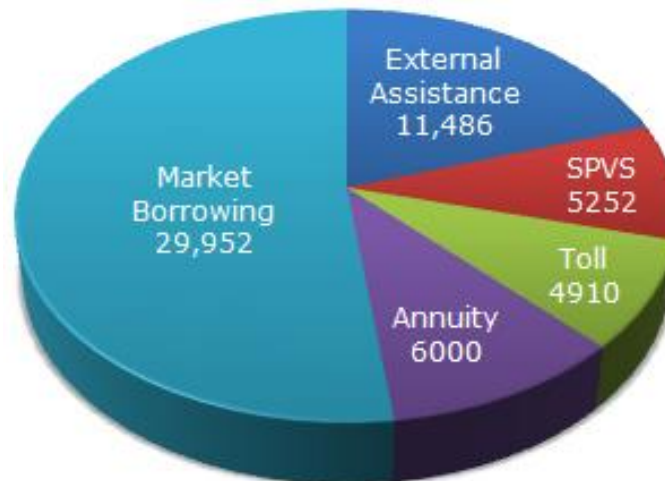
4. If 5500 copies are published and the transportation cost on them amounts to ₹.82500, then what should be the selling price of the book so that the publisher can earn a profit of 25%?

- a) ₹.180 b) ₹.187.5 c) ₹.142.75 d) ₹.193.5

5. Royalty on the book is less than the printing cost by:

- a)25% b)20% c)15% d)30%

Directions (6 – 10) : The following pie-chart shows the sources of funds to be collected by the National Highways Authority of India (NHAI) for its Phase II projects. Study the pie-chart and answers the question that follow. Sources of funds to be arranged by NHAI for Phase II projects (in crores ₹.) [Wipro]



6. Near about 20% of the funds are to be arranged through:

- a)SPVS b)External Assistance c)Annuity d)Market Borrowing

7. If NHAI could receive a total of ₹.9695 crores as External Assistance, by what percent (approximately) should it increase the Market Borrowing to arrange for the shortage of funds?

- a)20% b)15% c)10% d)6%

8. If the toll is to be collected through an outsourced agency by allowing a maximum 10% commission, how much amount should be permitted to be collected by the outsourced agency, so that the project is supported with Rs. 4910 crores?

- a)4401 b)3301 c)5401 d)6601

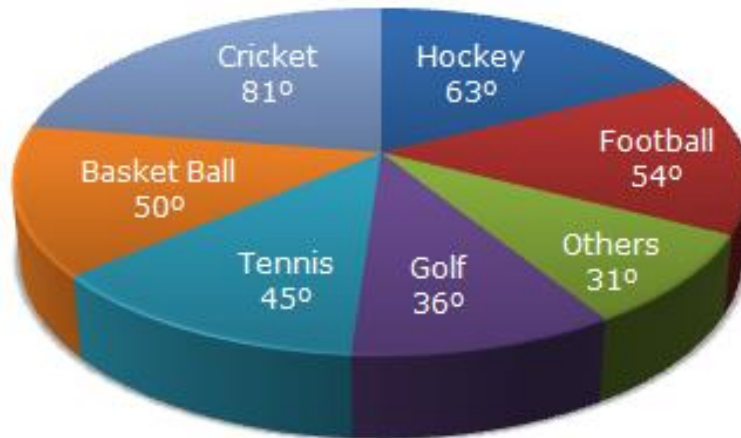
9. The central angle corresponding to Market Borrowing is

- a)187.2° b) 167.2° c) 197.2° d) 287.2°

10. The approximate ratio of the funds to be arranged through Toll and that through Market Borrowing is ?

- a)1 : 6 b)3 : 2 c)1 : 4 d)2 : 3

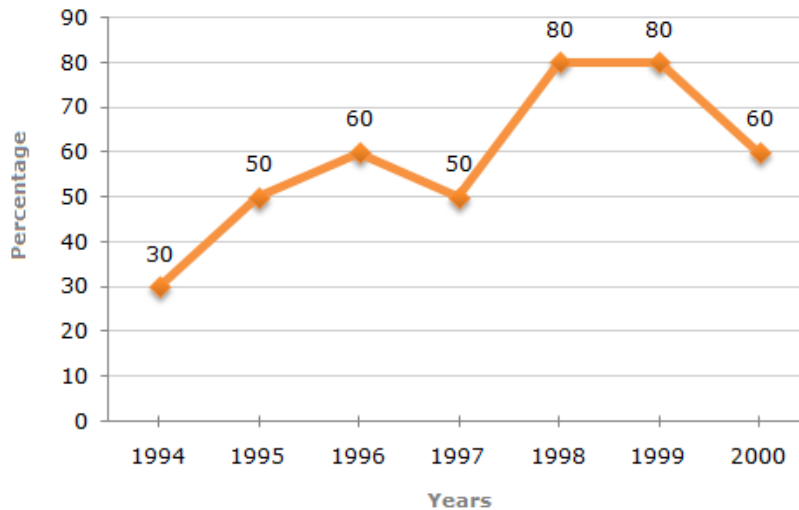
Directions (11-15) : The circle-graph given here shows the spending of a country on various sports during a particular year. Study the graph carefully and answer the questions given below it. [Amazon]



11. What percent of total spending is spent on Tennis?
a) $12\frac{1}{2}\%$ b) 10% c) $15\frac{1}{2}\%$ d) $14\frac{1}{2}\%$
12. How much percent more is spent on Hockey than that on Golf?
a) 20% b) 50% c) 75% d) 60%
13. If the total amount spent on sports during the year be ₹. 1,80,00,000 , the amount spent on Basketball exceeds on Tennis by:
a) ₹.250000 b) ₹.450000 c) ₹.200000 d) ₹.400000
14. How much percent less is spent on Football than that on Cricket?
a) 20% b) 25% c) $33\frac{1}{3}\%$ d) $66\frac{2}{3}\%$
15. If the total amount spent on sports during the year was ₹. 2 crores, the amount spent on Cricket and Hockey together was:
a) ₹.4000000 b) ₹.6000000 c) ₹.2000000 d) ₹.8000000

LINE GRAPH

Directions(1-5) : The following line graph gives the Percentage of Candidates Qualified to Appeared in an Examination Over the Years [EPAM]



1. The difference between the percentage of candidates qualified to appeared was maximum in which of the following pairs of years?

- a)1994 & 1995 b)1997 & 1998 c)1999 & 2000 d)1995 & 1996

2. In which pair of years was the number of candidates qualified, the same?

- a)1994 & 1995 b)1999 & 2000 c)1995 d)Data inadequate

3. If the number of candidates qualified in 1998 was 21200, what was the number of candidates appeared in 1998?

- a)26500 b)24500 c)28000 d)23600

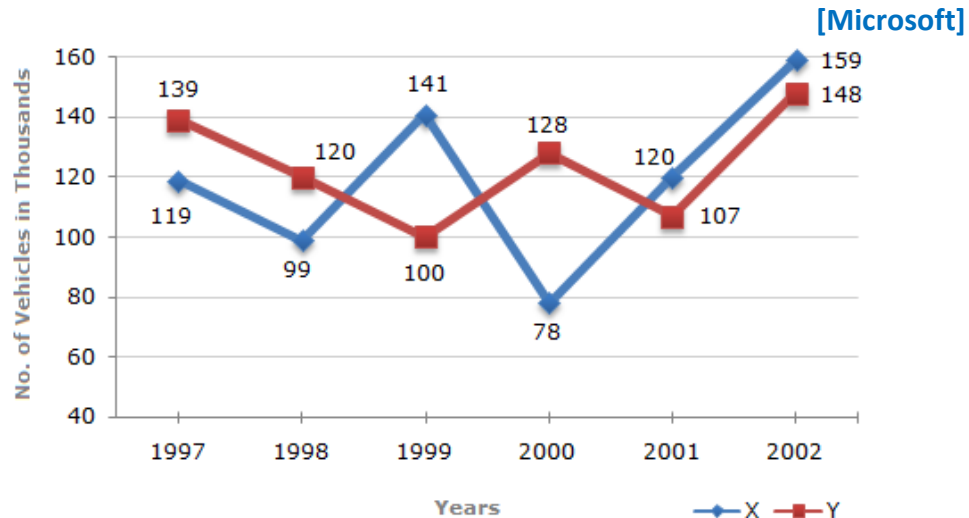
4. If the total number of candidates appeared in 1996 and 1997 together was 47400, then the total number of candidates qualified in these two years together was?

- a)37200 b)34000 c)39000 d)Data Inadequate

5. The total number of candidates qualified in 1999 and 2000 together was 33500 and the number of candidates appeared in 1999 was 26500. What was the number of candidates in 2000?

- a)20500 b)20800 c)20000 d)29600

Directions(6-10) : Study the following line graph and answer the questions based on it.
Number of Vehicles Manufactured by Two companies over the Years (Number in Thousands)



6. What is the difference between the number of vehicles manufactured by Company Y in 2000 and 2001 ?

- a)20000 b)22000 c)21000 d)24000

7. What is the difference between the total productions of the two Companies in the given years ?

- a)22000 b)26000 c)24000 d)28000

8. What is the average numbers of vehicles manufactured by Company X over the given period ? (rounded off to nearest integer)

- a)119333 b)139993 c)129999 d)239999

9. . In which of the following years, the difference between the productions of Companies X and Y was the maximum among the given years ?

- a)1997 b)1998 c)1999 d)2000

10. The production of Company Y in 2000 was approximately what percent of the production of Company X in the same year ?

- a)100% b)164% c)150% d)120%